University of Moratuwa
Faculty of Engineering
Department of Electronic & Telecommunication Engineering

EN2091: Laboratory Practice and Project

Team IonSplice

Wave Sync – RF Linked Alarm Clock

Group Members

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1. Introduction

1.1. Problem Description

In many real-world environments such as hostels, factories, laboratories, educational institutions, and large residential buildings, a single alarm clock is often insufficient to notify all intended users at the same time. Conventional alarm systems depend mainly on audible alerts, which are limited by physical distance, background noise, and structural barriers.

As a result, individuals located far from the alarm source may fail to receive the alert in time. This creates a requirement for a wireless alarm distribution system capable of transmitting an alarm signal to multiple locations simultaneously with minimal delay and high reliability.

1.2. Motivation

The primary motivation of this project is to design a simple, reliable, and low-cost wireless alarm system using purely analog electronics. By eliminating microcontrollers, software, and complex communication protocols, the system achieves ultra-low latency, improved reliability, and ease of maintenance.

The use of RF communication allows the alarm signal to be transmitted instantly to multiple receivers, making the system suitable for synchronized alerting applications.

1.3. Justification for Selection

This project was selected to demonstrate the effective application of core analog electronics and communication principles. The design integrates Digital-to-Analog Conversion (DAC), voltage comparison using comparators, pulse generation, and RF transmission using amplitude modulation.

The RF Linked Alarm Clock proves that practical engineering problems can be solved efficiently using analog techniques without relying on digital processing or embedded software, making it ideal for educational and low-cost industrial applications.

2. Technical Feasibility

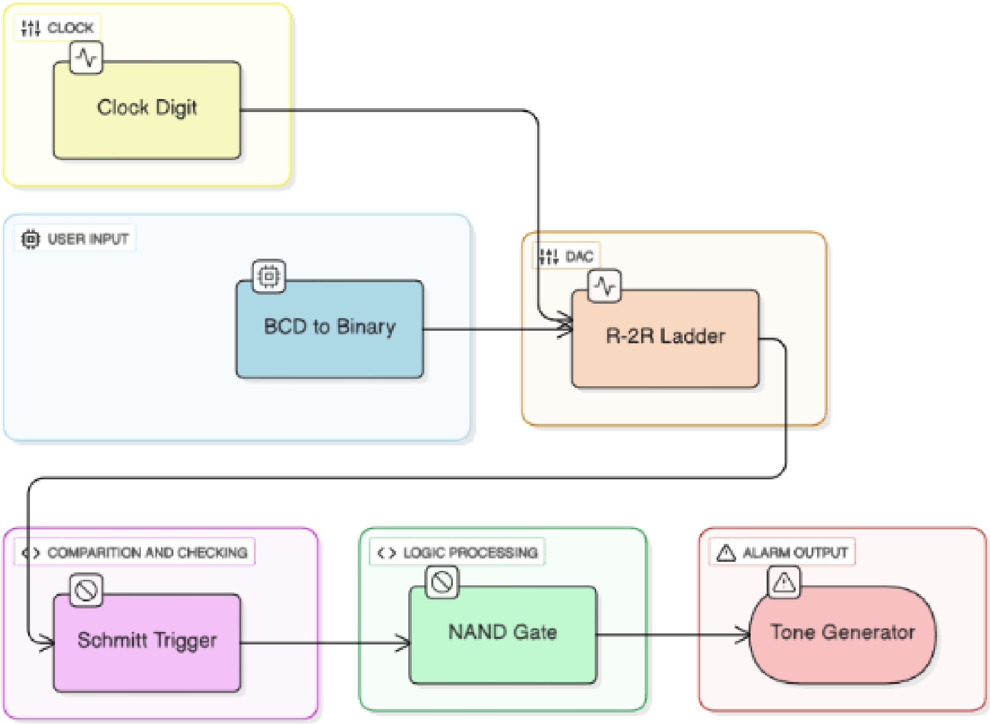
The RF Linked Alarm Clock is technically feasible as it uses readily available and well-established analog components.

- Digital clock digit bits are processed without any software control.
- A DAC converts digital time data into a corresponding analog voltage.
- A comparator accurately detects alarm time matching.
- 433 MHz RF transmitter and receiver modules provide stable wireless communication.
- AM-based On-Off Keying (OOK) modulation ensures simple and robust transmission.
- Total power consumption remains below 1 W, enabling continuous operation.

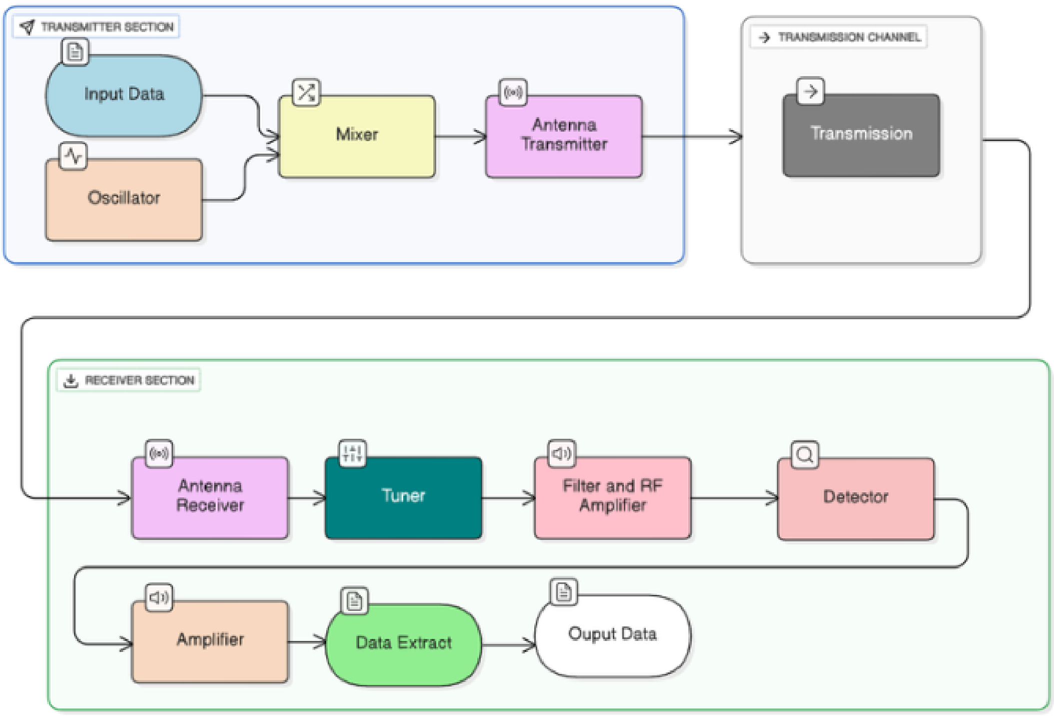
All components operate within safe voltage and current limits, ensuring system stability and reliability.

3. Product Architecture

3.1. Block Diagram



Alarm Triggering



RF Transmission

3.2. Description of Blocks and Circuits

1. Clock Digit Block

- Function: Generates the time information as discrete digit bits representing hours, minutes, and seconds.
- Circuit Used: Custom analog clock circuit.

2. User Input / Encoder

- Function: Allows the user to set the alarm time by converting the alarm setting into binary form.
- Component Used: Manual rotary encoder with BCD to binary conversion logic.

3. DAC (R – 2R Ladder)

- Function: Converts the digital alarm setting and clock digit bits into an analog voltage for comparison
- Circuit Used: R-2R ladder network.

4. Differential Amplifier

- Function: Computes the difference between the clock voltage and the user-set alarm voltage from the DAC, producing a voltage proportional to how close the clock is to the alarm.
- Circuit Used: Op-amp configured as a differential amplifier.

5. Modulus (Absolute Value) Circuit

- Function: Converts the differential voltage into its absolute value, ensuring that the signal represents the magnitude of the difference regardless of polarity.
- Circuit Used: Op-amp-based modulus circuit.

6. Logic Processing (NAND Gate)

- Function: Receives the modulus signal and produces a clean digital trigger pulse when the alarm condition is met.
- Implementation: Realized using NPN transistors arranged such that the output is LOW only when all inputs are HIGH; otherwise, output is HIGH.
- Component Used: NPN transistors, base resistors, and a collector pull-up resistor.

7. Alarm Output (Tone Generator / Buzzer)

- Function: Produces an alert when the alarm trigger pulse is generated.
- Component Used: LED

8. Transmitter Section

- Function: Converts the alarm trigger pulse into an RF signal for wireless transmission.
- Component Used: Mixer, oscillator (433 MHz), and antenna.

9. Receiver Section

- Function: Captures the transmitted RF signal and recovers the alarm trigger for local activation.
- Component Used: Antenna, tuner, filter & amplifier, detector, and output logic to drive the local buzzer.

3.3. System Operation (Two-Stage Analog Design)

Stage 1 – Alarm Triggering

The digital clock continuously outputs digit bits representing the current time. These bits are converted into an analog voltage using a DAC. A comparator compares this voltage with the user-set alarm voltage. When both voltages become equal, a trigger pulse is generated.

Stage 2 – RF Transmission

The trigger pulse enables the RF transmitter operating at 433 MHz. Using AM modulation with On-Off Keying, the alarm signal is broadcast wirelessly. All receivers tuned to the same frequency receive the signal simultaneously and activate their alarm outputs.

4. Final Product Details

4.1. Product Overview

The RF Linked Alarm Clock is an analog wireless alarm distribution system designed to broadcast an alarm signal from a single transmitter to multiple receivers. The system offers synchronized alerting with ultra-low latency and high reliability.

4.2. Key Technical Specifications

- Operating Frequency: 433 MHz (ISM Band)
- Modulation Technique: AM – On-Off Keying (OOK)
- Nominal RF Range: Up to 100 m
- Power Consumption: Less than 1 W

- Communication Type: One-to-Many Analog Broadcast
- Scalability: Unlimited Receivers

4.3.Enclosure Design

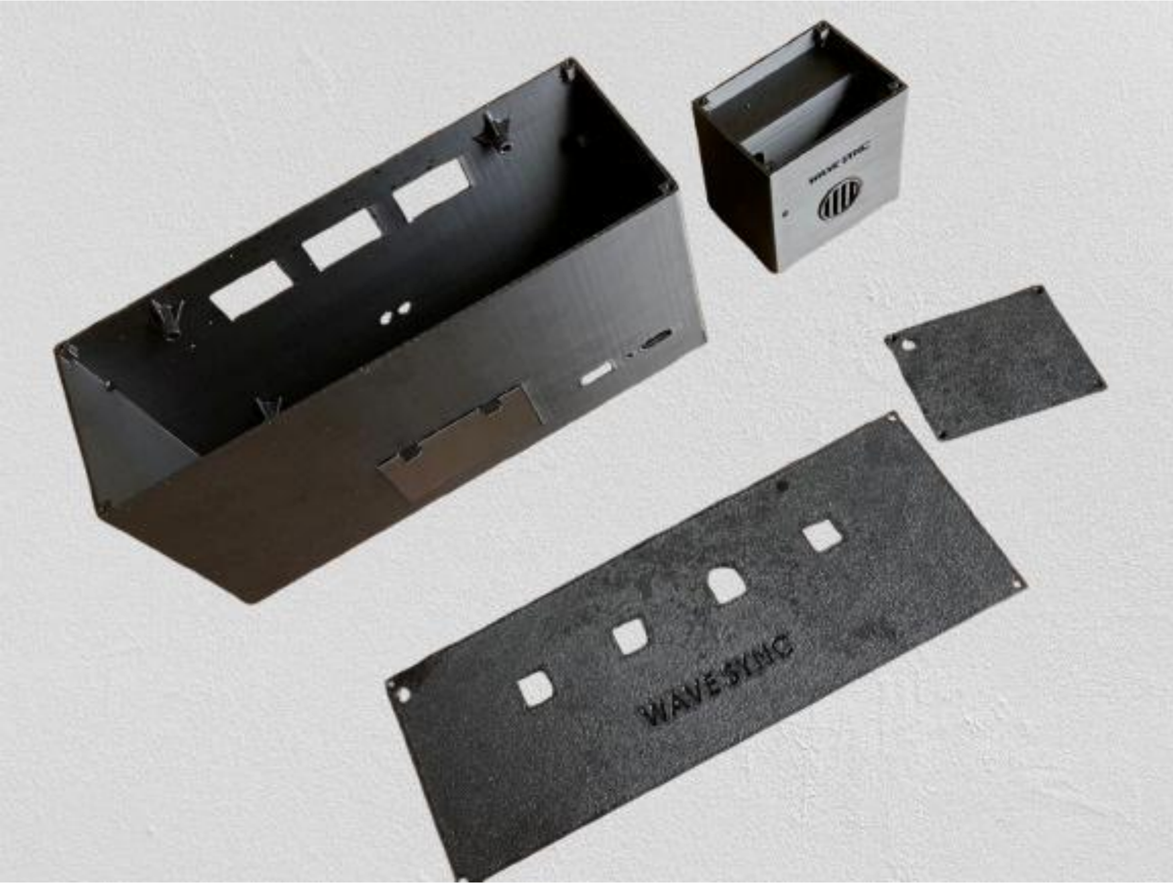
For the RF Linked Alarm Clock project, a compact and function-oriented enclosure was designed to accommodate the complete clock circuitry along with the DAC, comparator, pulse-shaping network, and RF transmission stage. The enclosure dimensions were selected to provide sufficient internal volume for discrete components, PCB routing, and interconnections while maintaining a small physical footprint suitable for stationary operation on a desk or wall mounting.

The internal layout is optimized to electrically and physically separate sensitive analog blocks. The analog clock generation circuit, including the oscillator and timing components, is positioned away from the RF transmission section to minimize electromagnetic interference and preserve stable frequency operation. Careful grounding and decoupling strategies were implemented to reduce noise coupling between the comparator stage, DAC, and RF components, ensuring reliable detection of alarm threshold events.

User interaction elements are strategically placed: the clock display LED is located on the front face for easy, continuous monitoring of the current time and alarm status, while the manual encoder for adjusting the alarm reference voltage is positioned on the top panel for intuitive access. This analog alarm-setting method eliminates the need for digital processing while providing precise, continuous adjustment capability. Visual indicators such as LEDs and an audible buzzer are positioned for clear feedback when the alarm activates.

Battery power is supplied through a dedicated compartment with a secure connector, ensuring portability and eliminating dependency on mains power. The enclosure includes openings for safe battery insertion and replacement while maintaining electrical isolation from sensitive analog circuitry.

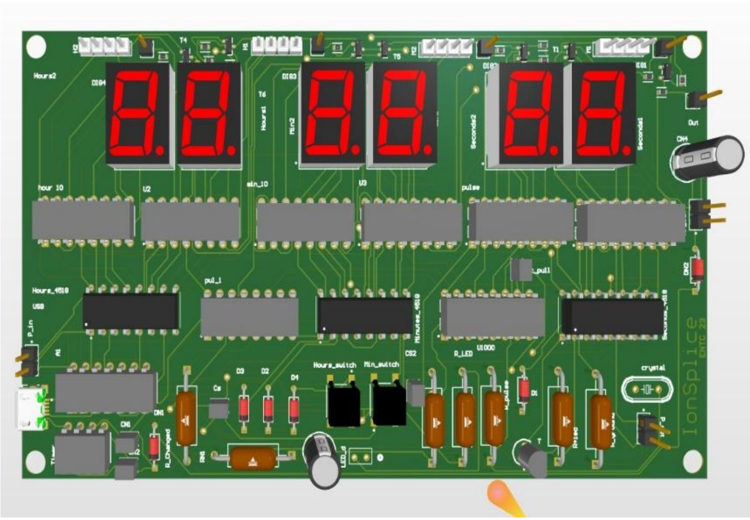
Plastic was selected as the enclosure material due to **its** lightweight, durable, and electrically insulating properties, which are ideal for analog circuits. This choice supports low-cost prototyping, easy fabrication, and potential scalability for mass production. The overall design balances functionality, noise immunity, accessibility, and durability, providing a stable and user-friendly platform for the complete RF-linked analog alarm system.

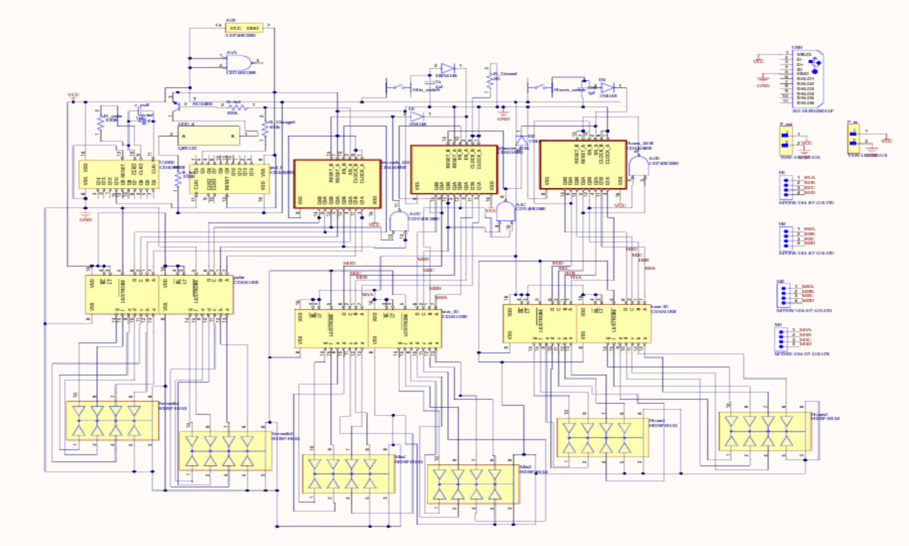


Parts of the Enclosure

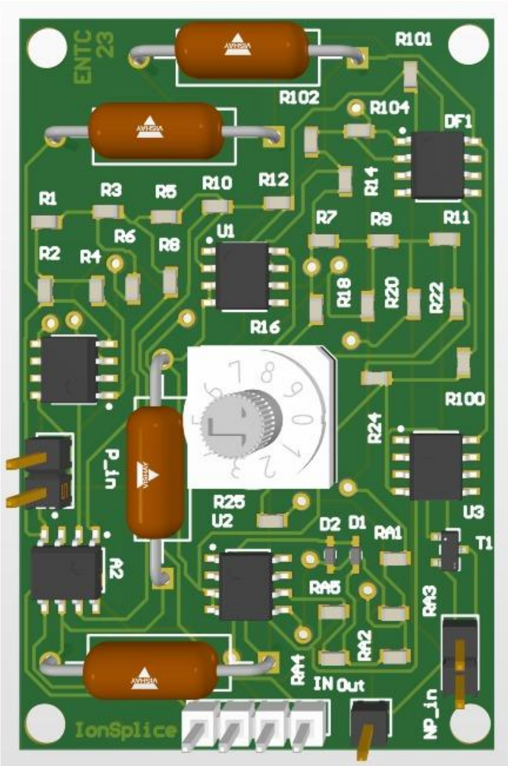
Final Model of the Enclosure

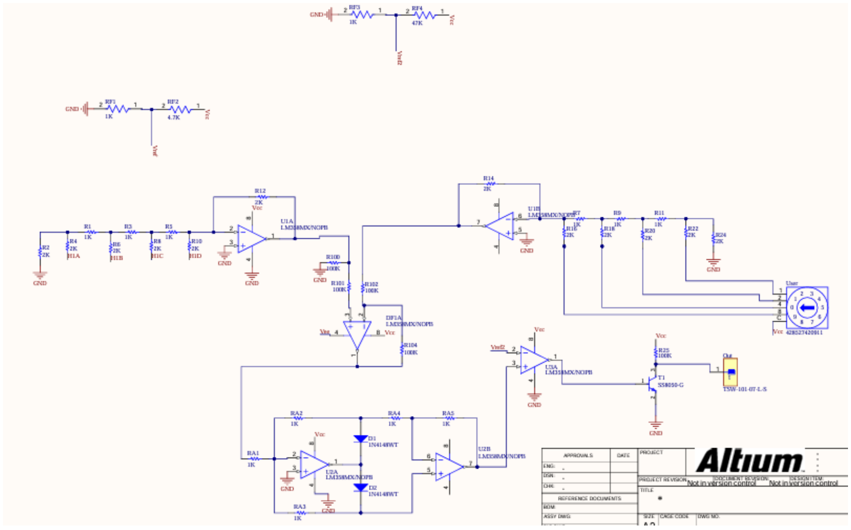
4.4.PCB Design



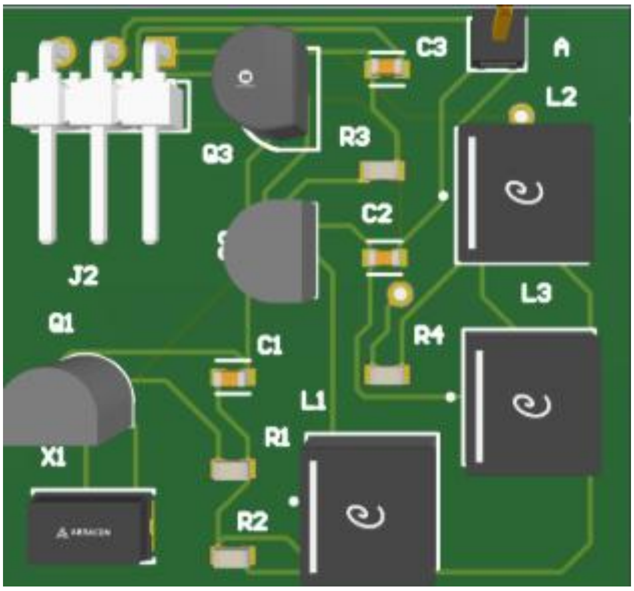


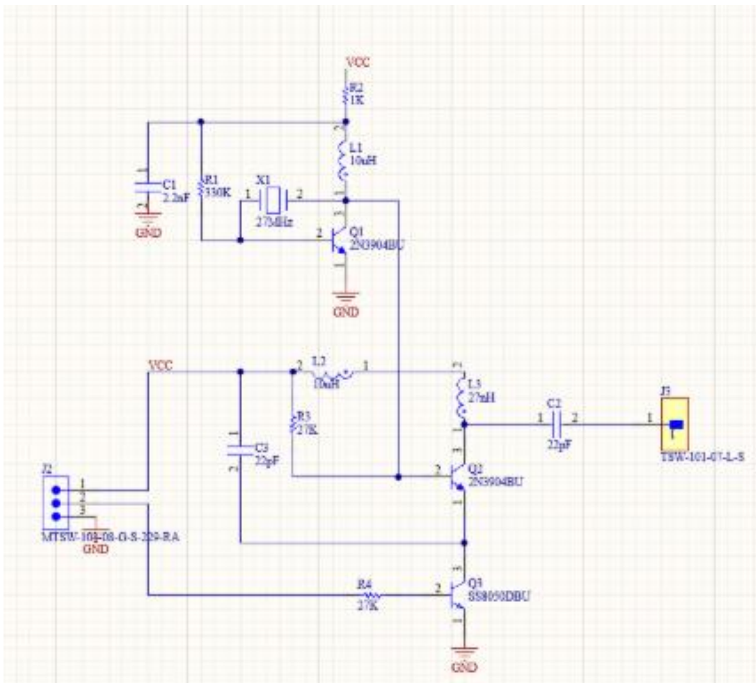
Clock



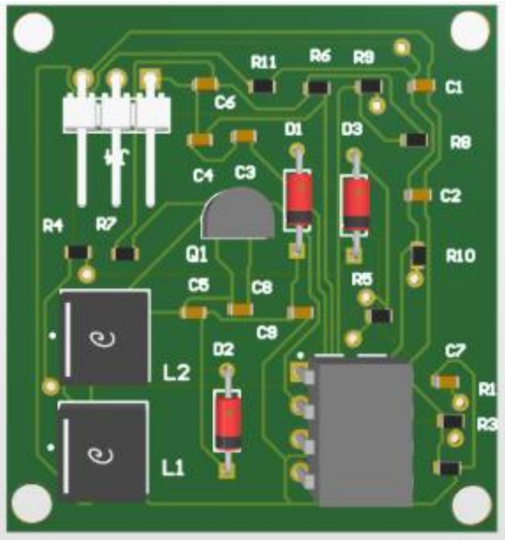


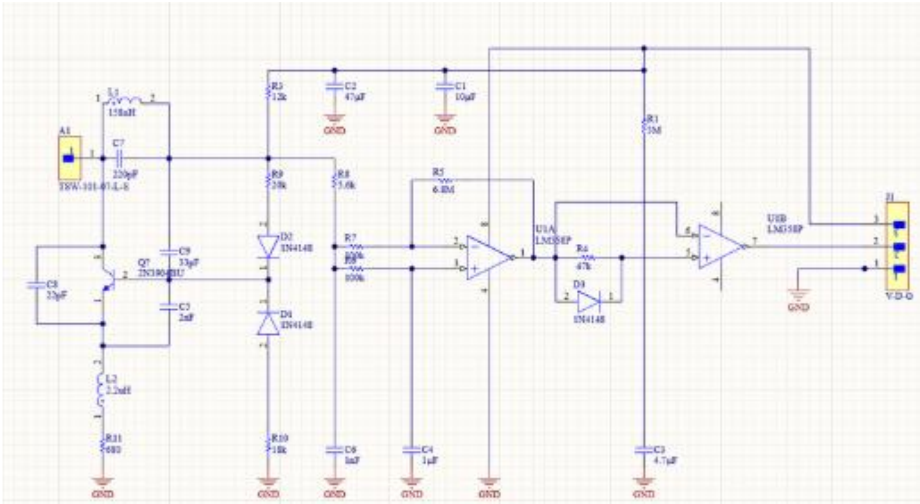
DAC & Comparator





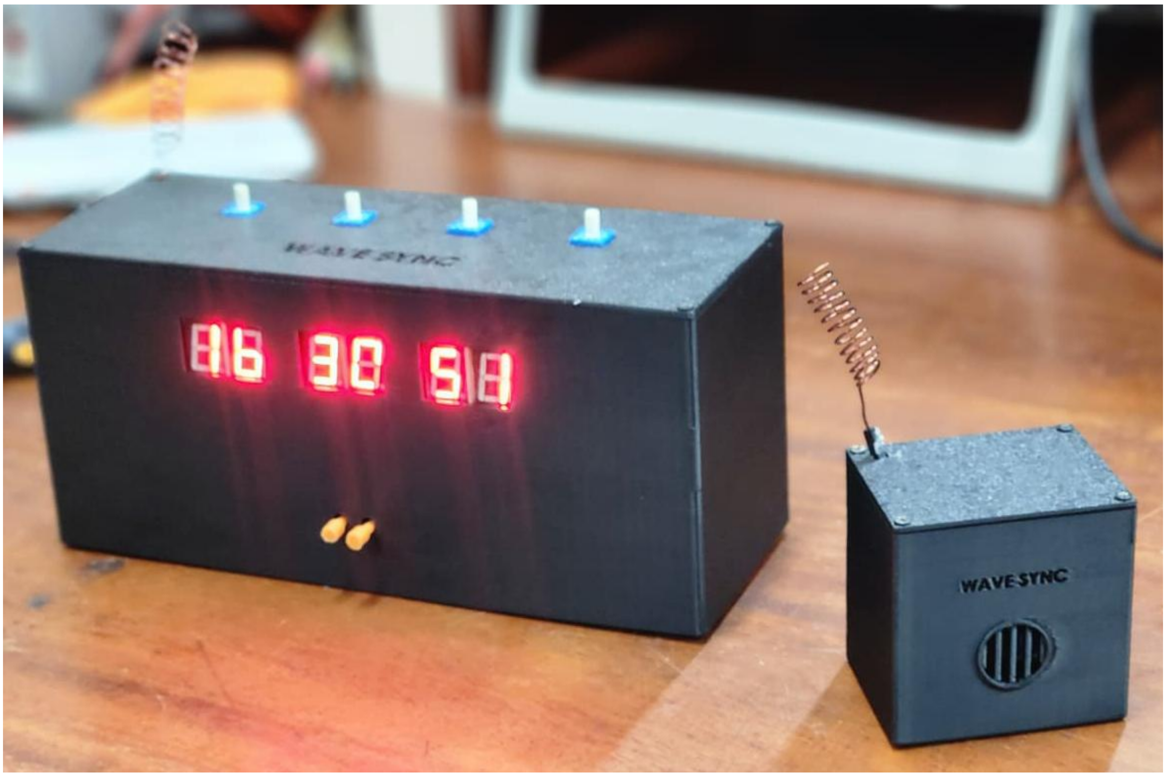
Transmitter





Receiver

4.5.Final Product



Final Product

5. Future Improvements

- Push to talk
- Visual alert modules for hearing-impaired users
- Integrate low-power or sleep-mode functionality to extend battery life.

6. Task Allocation

- System Concept and Architecture – Team
- Clock Circuit – Umair A.
- DAC, and Comparator Design – Ahamed A.M.S.
- Transmitter and Receiver Circuit – Hakam M.R.A.
- PCB Layout and Design – Hakam M.R.A. , Ahamed A.M.S. & Umair A.
- Enclosure Design and Development – Manatunga K.D.